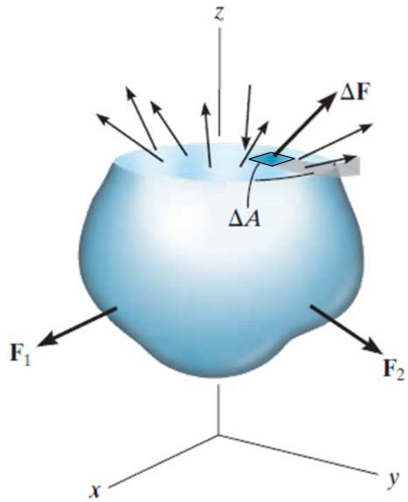
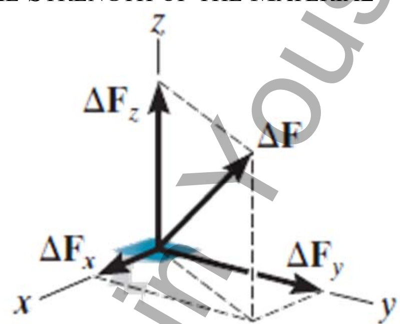
Stress

CONCEPT: STRESS IS ASSOCIATED WITH THE STRENGTH OF THE MATERIAL





ΔF is the intensity of the internal force acting on a specific plane (area) passing through a point

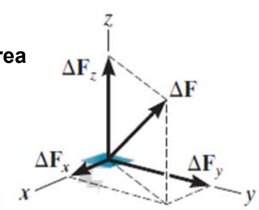
Stress

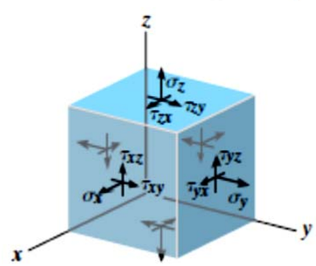
CONCEPT

Stress represents a force per unit area

Normal Stress $$\sigma_z = \lim_{\Delta A \to 0} \frac{\Delta F_z}{\Delta A}$$

Shear Stress
$$\tau_{zx} = \lim_{\Delta A \to 0} \frac{\Delta F_x}{\Delta A}$$
$$\tau_{zy} = \lim_{\Delta A \to 0} \frac{\Delta F_y}{\Delta A}$$



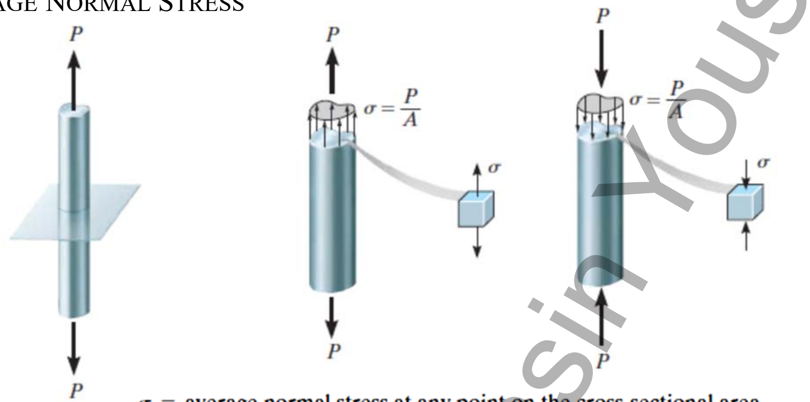


subscript notation z specifies the orientation of the area

Units: $Pa = N/m^2$

Stress

Average Normal Stress



$$\sigma = \frac{P}{A}$$

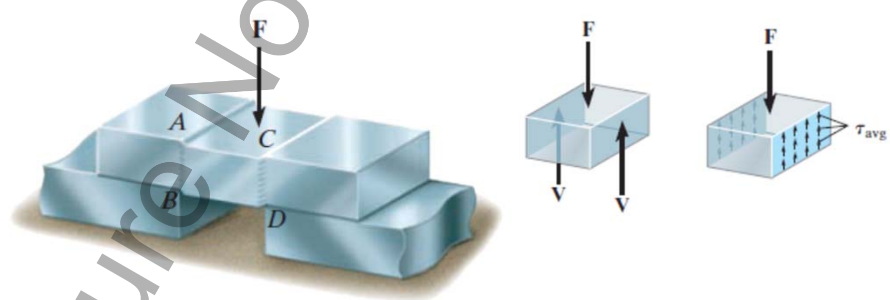
σ = average normal stress at any point on the cross-sectional area

P = *internal resultant normal force*, which acts through the *centroid* of the cross-sectional area. P is determined using the method of sections and the equations of equilibrium

A = cross-sectional area of the bar where σ is determined

Stress

Average Shear Stress



$$\tau_{avg} = \frac{V}{A}$$

τ_{avg} = average shear stress at the section, which is assumed to be the *same* at each point located on the section

V = internal resultant shear force on the section determined from the equations of equilibrium

A = area at the section

Stress

Allowable Stress

$$\textit{Factor of Safety}\ (F.S.) = \frac{\textit{Failure Load}}{\textit{Allowable Load}}$$

$$\text{F.S.} = \frac{F_{\text{fail}}}{F_{\text{allow}}}$$

$$\text{F.S.} = \frac{\sigma_{\text{fail}}}{\sigma_{\text{allow}}}$$

$$\text{F.S.} = \frac{\tau_{\text{fail}}}{\tau_{\text{allow}}}$$

Generally $1 < F.S < 3$

Strain

Introduction

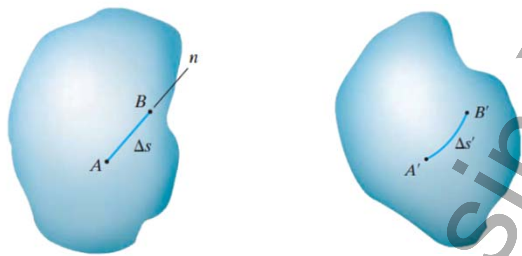
Before Deformation

After Deformation

Strain

NORMAL STRAIN

normal strain as the change in length of a line per unit length

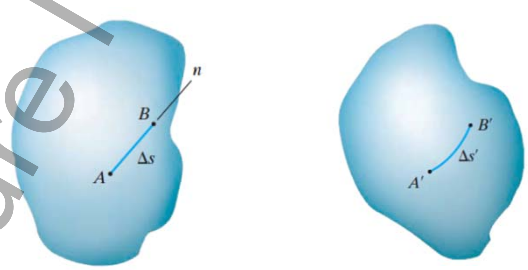


$$\epsilon_{avg} = \frac{\Delta s' - \Delta s}{\Delta s}$$

Strain

NORMAL STRAIN

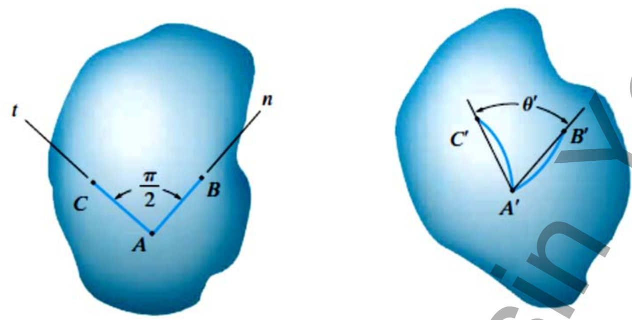
normal strain as the change in length of a line per unit length



$$\epsilon = \lim_{B \to A \text{ along } n} \frac{\Delta s' - \Delta s}{\Delta s}$$

Strain

SHEAR STRAIN



Undeformed body

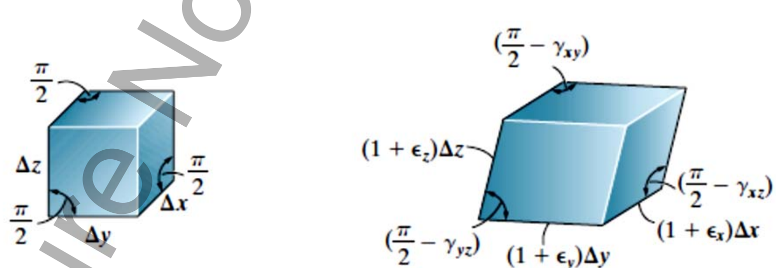
Deformed body

$$\boxed{\gamma_{nt} = \frac{\pi}{2} - \lim_{\substack{B \to A \text{ along } n \\ C \to A \text{ along } t}} \theta'}$$

Mechanics Of Materials Basic Mech. Engg. | Mechanics Of Materials | Loads | Stress | Strain

Strain

SHEAR STRAIN



Undeformed element

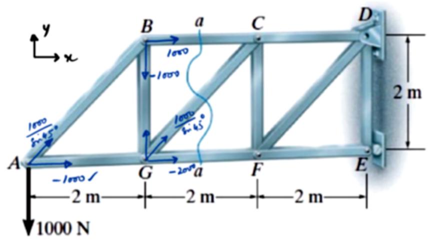
(b)

Deformed element

(c)

Factor of Safety

EXAMPLE



Lead UTS 12 MPa
Al " 40 MPa
IRON 5 MPa

$F_{GF} = -2000\ N$

INDUCED STRESS IN LINK GF $= \frac{F}{A} = \frac{2000}{0.0004} = 5M\ \frac{N}{m^2} = 5M\ Pa$

What should be the metal?

UTS material > Induced stress

FACTOR OF SAFTY $FS = \frac{UTS}{IS} =$

IRON	LEAD	AL
$\frac{5}{5} = 1$	$\frac{12}{5} = 2.4$	$\frac{40}{5}$ 8

$FS = 4$ AL

What should be the size?

$IS = \frac{UTS}{FS} = \frac{40M}{4} = 10\ M$

$\frac{2000}{10M} = A \Rightarrow A = 0.0002$

Side = 0.0142 m = 14.2 mm

$\frac{F}{A} = 10$

CROSS SECTION AREA

20mm = 0.02 m

$A = 0.0004\ m^2$